

High-precision randomized iterative methods for the random feature method

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1 Introduction

The Random Feature Method (RFM) has emerged as an effective approach for solving partial differential equations (PDEs), bridging traditional numerical methods and machine learning-based algorithms. While RFM offers significant advantages in handling complex geometries with exponential convergence rates, its practical implementation faces a critical challenge: solving large-scale, ill-conditioned, and overdetermined sparse least squares problems.

Traditional solvers for these problems fall into two categories: direct methods (such as SVD and QR decomposition) that achieve high accuracy but with computational complexity of $O(mn \min(m, n))$, and iterative methods with per-iteration complexity of $O(mn)$. For RFM applications, where $m \gg n$ by several orders of magnitude, direct methods become computationally expensive, while classical iterative methods often fail to achieve high precision due to the ill-conditioning.

This paper addresses this challenge by introducing two high-precision randomized iterative methods: Count Sketch QR Preconditioned LSQR (CSQRP-LSQR) and Count Sketch SVD Preconditioned LSQR (CSSVDP-LSQR). These methods combine randomized sketching techniques with preconditioning strategies to transform ill-conditioned problems into well-conditioned ones, enabling efficient and accurate solutions.

2 Algorithm Details

2.1 Core Components

Both proposed methods are built upon three key components:

1. **Count Sketch Technique:** A dimensionality reduction method that preserves the essential information of the original matrix while significantly reducing computational requirements.
2. **Preconditioner Construction:** Using the sketched matrix to construct a preconditioner through either QR factorization or singular value decomposition (SVD).
3. **Iterative Solver:** Applying the LSQR method to the preconditioned system.

2.2 Count Sketch Transform

The count sketch transform is defined as $S = \Phi D \in \mathbb{R}^{s \times m}$, where:

- D is an $m \times m$ random diagonal matrix with each diagonal entry independently chosen to be +1 or -1 with equal probability
- $\Phi \in \{0, 1\}^{s \times m}$ is a binary matrix with $\Phi_{h(i), i} = 1$ and all remaining entries 0
- $h : [m] \rightarrow [s]$ is a random map such that for each $i \in [m]$, $h(i) = j$ with probability $1/s$ for each $j \in [s]$

This technique provides an ℓ_2 -subspace embedding for the column space of a matrix A , meaning for all $x \in \mathbb{R}^n$:

$$(1 - \epsilon)\|Ax\|_2^2 \leq \|SAx\|_2^2 \leq (1 + \epsilon)\|Ax\|_2^2 \quad (1)$$

The count sketch method offers optimal computational complexity of $O(\text{nnz}(A))$ (where nnz denotes the number of non-zero elements) and requires only one non-zero entry per column, preserving the sparsity structure of the original matrix.

2.3 CSQRP-LSQR Algorithm

The CSQRP-LSQR method proceeds as follows:

1. Choose an oversampling factor $\gamma > 1$ and set $s = \gamma n$ (with $s < m$)
2. Generate an $s \times m$ count sketch matrix S and compute $\tilde{A} = SA$
3. Apply QR factorization to $\tilde{A}$ to obtain the upper triangular matrix R

4. Compute the preconditioned matrix $B = AR^{-1}$
5. Use the LSQR method to find an approximate solution $y^{(k)}$ of $\min_{y \in \mathbb{R}^n} \|By - b\|_2$
6. Return $x^{(k)} = R^{-1}y^{(k)}$

2.4 CSSVDP-LSQR Algorithm

The CSSVDP-LSQR method is similar but uses SVD instead of QR factorization:

1. Choose an oversampling factor $\gamma > 1$ and set $s = \gamma n$ (with $s < m$)
2. Generate an $s \times m$ count sketch matrix S and compute $\tilde{A} = SA$
3. Compute the compact SVD $U\Sigma V^T$ of matrix $\tilde{A}$, where $r = \text{rank}(\tilde{A})$, $U \in \mathbb{R}^{s \times r}$, $\Sigma \in \mathbb{R}^{r \times r}$, $V \in \mathbb{R}^{n \times r}$, and let $P = V\Sigma^{-1}$
4. Compute the preconditioned matrix $B = AP$
5. Use the LSQR method to compute an approximate solution $y^{(k)}$ for $\min_{y \in \mathbb{R}^n} \|By - b\|_2$
6. Return $x^{(k)} = Py^{(k)}$

The CSSVDP-LSQR method is particularly effective for matrices that are rank-deficient or have very high condition numbers, as it performs a truncated SVD that effectively filters out numerically insignificant singular values.

3 Theoretical Findings

3.1 Condition Number Bounds

A key theoretical finding is that the condition number of the preconditioned matrix is bounded independently of the condition number of the original matrix:

Let matrix A in the least squares problem be of full column rank and the construction of the count sketch matrix S satisfy certain conditions. Then, with probability at least $1 - \delta$:

- The upper triangular matrix R obtained from the QR decomposition of $\tilde{A}$ is invertible

- The condition number of the preconditioned matrix AR^{-1} satisfies $\kappa(AR^{-1}) \leq \sqrt{\frac{1+\epsilon}{1-\epsilon}}$
- The preconditioned least squares problem has a unique solution

This result shows that the proposed method can transform an ill-conditioned problem into a well-conditioned one with high probability, with the new condition number bounded by a constant that depends only on ϵ .

3.2 Error Estimates

For the CSQRP-LSQR method, the paper provides detailed error estimates:

Under the same conditions as above, and letting x^* and y^* be the least squares solutions to the original and preconditioned problems respectively, given a tolerance $\tau > 0$, when the LSQR method is used to solve the preconditioned system, if the number of iteration steps satisfies:

$$k \geq \frac{\ln(2) + |\ln(\tau)|}{|\ln(\epsilon)|} \quad (2)$$

then:

$$\|y^{(k)} - y^*\|_{B^T B} \leq \tau \|y^*\|_{B^T B} \quad (3)$$

This implies that the error in the original problem satisfies:

$$\|x^{(k)} - x^*\|_{A^T A} \leq \tau \|x^*\|_{A^T A} \quad (4)$$

The paper also provides error estimates for the CSSVDP-LSQR method when applied to approximately rank-deficient problems, showing that under certain conditions, the method can determine the effective rank correctly and achieve high precision.

3.3 Convergence Rate Analysis

The paper shows that the convergence rate of LSQR on the preconditioned system is determined by:

$$\frac{\|y^{(k)} - y^*\|_{B^T B}}{\|y^{(0)} - y^*\|_{B^T B}} \leq 2 \left(\frac{\sqrt{\kappa(B^T B)} - 1}{\sqrt{\kappa(B^T B)} + 1} \right)^k \quad (5)$$

Since $\kappa(B^T B) \leq \frac{1+\epsilon}{1-\epsilon}$, this leads to exponential convergence with rate bounded by ϵ , enabling high precision with relatively few iterations.

4 Numerical Examples and Results

4.1 Two-Dimensional PDEs

The paper presents extensive numerical results for various PDEs, including:

- **Timoshenko Beam Problem:** Comparing various methods, CSSVDP-LSQR achieved relative errors of 10^{-17} while CSQRP-LSQR reached 10^{-13} . In contrast, standard iterative methods like LSQR and LSMR could only achieve 10^{-4} after 10,000 iterations.
- **Stokes Flow Problem:** For a square domain with three holes, CSSVDP-LSQR achieved errors of 10^{-8} in u , 10^{-9} in v , and 10^{-5} in pressure p , significantly better than other methods.
- **Elasticity Problem with Complex Geometry:** Both CSQRP-LSQR and CSSVDP-LSQR methods achieved relative errors around 10^{-11} with errors in physical variables around 10^{-5} .

A key finding was that using a count sketch matrix with $s = 3n$ (i.e., $\gamma = 3$) generally yielded optimal total time and accuracy across problems, despite theoretical bounds suggesting $s = O(n^2)$.

4.2 Three-Dimensional PDEs

For three-dimensional problems, the methods demonstrated similar effectiveness:

- **Poisson Equation:** Both CSQRP-LSQR and CSSVDP-LSQR achieved relative errors down to 10^{-14} as problem size increased.
- **Helmholtz Equation:** For low-frequency solutions, CSQRP-LSQR achieved errors down to 10^{-14} , while for high and mixed-frequency problems, it reached 10^{-12} .
- **Elasticity Problem:** CSQRP-LSQR achieved relative L^2 errors down to 10^{-15} , outperforming direct methods in computational time for large problems.
- **Stokes Flow:** For a cubic domain with inhomogeneous boundary conditions, CSQRP-LSQR reached errors of 10^{-16} in 63 iterations.

4.3 Problems with Rank Deficiency or Infinite Condition Numbers

The paper evaluated CSSVDP-LSQR on matrices from the Florida Sparse Matrix Collection with rank deficiency or infinite condition numbers:

- CSSVDP-LSQR consistently achieved relative errors around 10^{-15} in 24-40 iterations.
- The condition number of preconditioned matrices AP ranged from 3 to 6, compared to original condition numbers reaching 10^{43} or infinity.
- CSQRP-LSQR was less effective for rank-deficient problems but still worked for some cases with condition numbers as high as 10^{15} .

The paper observed that singular values of preconditioned matrices were more closely clustered near 1, explaining the improved convergence behavior.

5 Implications and Advantages

5.1 Computational Efficiency

The proposed methods offer several computational advantages:

- CSQRP-LSQR is computationally faster on large-scale problems compared to the state-of-the-art sparse direct solver SPQR.
- The count sketch technique preserves sparsity patterns, allowing efficient operations without explicitly forming the sketching matrix.
- As problem size increases, the advantage of these methods over direct solvers becomes more pronounced.

5.2 Accuracy and Reliability

The accuracy achieved is comparable to direct methods:

- The relative least squares error precision matches that of SPQR and Householder QR methods.
- The methods handle problems with condition numbers up to 10^{20} while maintaining high precision.
- CSSVDP-LSQR successfully solves problems with rank deficiency or infinite condition numbers where direct methods fail.

5.3 Practical Implications for RFM

These methods significantly enhance the applicability of RFM for solving PDEs:

- They enable high-precision solutions necessary for RFM to achieve its theoretical exponential convergence rate.
- They make RFM practical for complex geometries and high-complexity solutions.
- The enhanced solver performance allows RFM to compete with both traditional numerical methods and machine learning approaches.

The paper notes that while theory suggests a sketching dimension of $s = O(n^2)$ is required, numerical experiments show that $s = 3n$ works well in practice, significantly improving the practical utility of count sketch transformations.

5.4 Future Research Directions

The work opens several promising research directions:

- Closing the gap between theoretical requirements ($s = O(n^2)$) and practical performance ($s = O(n)$) for count sketch dimensions.
- Further optimizing the preconditioning strategies for specific problem structures.
- Exploring parallel implementations to further enhance computational efficiency.
- Applying the methods to even broader classes of PDEs and other scientific computing problems.